Class 10

Productivity and Wages

Part II — Building the American Growth Machine

The American Economy — draft course materials (proposal under discussion)

The point of today's class

Productivity growth is the central reason workers can become richer over time — but the link from productivity to pay depends on what we mean by "pay," which workers we mean, what prices they face, and what institutions govern labor markets.

Why it matters

- Would you rather be a typical worker in 1820, 1920, 1970, or today? Ordinary workers today are vastly richer — that is the big fact.
- Two bad stories to reject:
  - "Capitalism just exploits workers; productivity rises but workers get nothing."
  - "Productivity automatically raises everyone's wages immediately."
- Better economics: productivity creates the *possibility* of higher pay; markets and institutions determine how it is shared.
- This class cashes out the Part II arc: ideas (8), firms and scale (9), and now — who gains?

Today

1. Why productivity is the wage engine
2. What counts as "pay"?
3. The productivity–pay link, carefully
4. Workers as a group vs. the typical worker
5. Four eras of American wages
6. Measurement clinic: reading famous charts critically

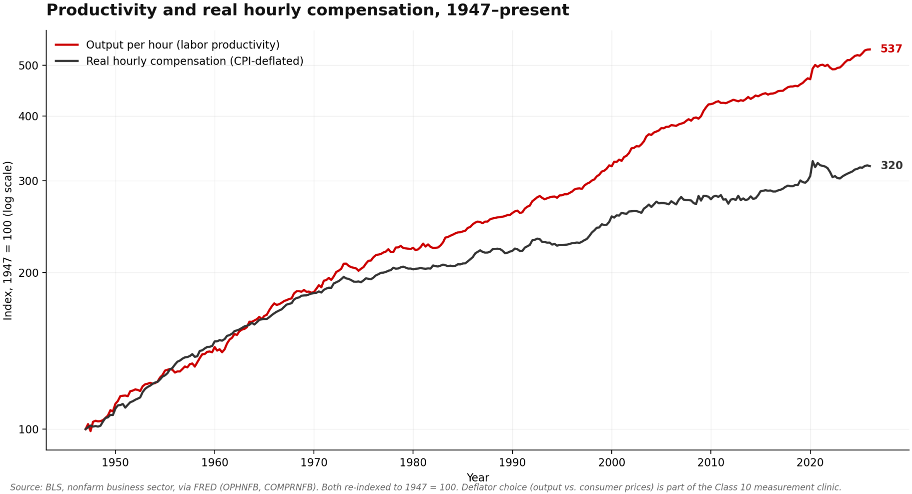
Productivity is why wages *can* rise

- The labor-demand logic in words: firms hire up to the point where the cost of one more worker equals the value of what that worker adds ($w \approx \text{MRPL}$).
- So higher productivity raises the **demand for labor**.
- If workers produce more per hour, society can afford higher pay, lower prices, more profits, more leisure, or more public goods.
- Productivity is the possibility — not the guarantee.

What counts as "pay"?

- BLS, December 2025: private-industry compensation costs \$46.15 per hour worked — \$32.36 in wages and salaries, \$13.79 in benefits.
- Wages alone miss health insurance, retirement, paid leave, and legally required benefits.
- Classroom question: a \$5,000 health-insurance benefit instead of a \$5,000 raise — are you better paid? Economically yes, though it may not feel that way.

Productivity and real hourly compensation, 1947–present



Source: Bureau of Labor Statistics, nonfarm business sector, via FRED (OPHNFB, COMPRNFB), both re-indexed to 1947 = 100. The deflator choice is part of the measurement clinic.

The link is real — but not mechanical (Stansbury–Summers)

- 1973–2016: one percentage point of extra productivity growth is associated with 0.7 to 1 point higher median *and* average compensation growth.
- Productivity still matters for pay — but other forces suppressed typical compensation over the same decades.
- Not broken, not automatic: exactly what the labor-demand logic predicts once supply, skills, bargaining, and prices enter.

Workers as a group vs. the typical worker

- **Labor share**: workers' total compensation as a share of output — labor vs. capital. It cannot show inequality *among* workers.
- **Mean vs. median**: the average worker and the typical worker are not the same person.

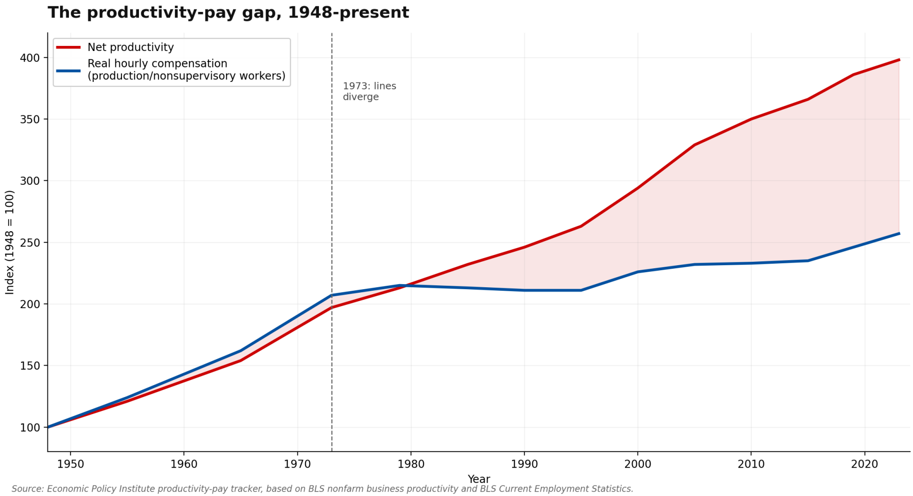
Economy A	Economy B
All workers get 2% raises	Top workers get 10%, others 0%
Average rises	Average rises
Median rises	Median may not rise

Four eras of American wages

Era	Wage story	Mechanism
1800s	rising long-run wages, harsh conditions	land, industrialization, immigration
1940s–1960s	broad compression, rising middle class	education, unions, norms, productivity
1970s–1990s	slower median growth, rising inequality	skills, globalization, deunionization
1990s–today	mixed: innovation, inequality, tight markets	skills, firms, places, policy

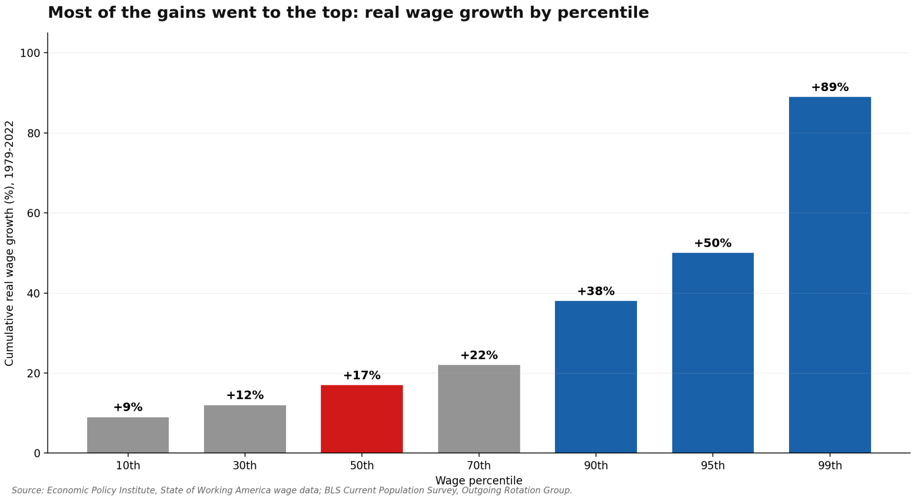
- Growth is transformative over generations but not automatically humane in the short run (the Mill–Engels caveat; Crafts on the Industrial Revolution).
- Mid-century: the "Great Compression" (Goldin–Margo). Post-1980 dispersion: the supply and demand for skills (Autor).

The productivity–pay gap: a chart to read critically



Source: Economic Policy Institute. Used as a measurement clinic: wages vs. total compensation, average vs. median, output vs. consumer deflators.

Real wage growth by percentile



Source: Economic Policy Institute, State of Working America.

Measurement clinic: what choices built these charts?

Not "is the chart a lie?" but: what does each choice do?

- **Wages vs. total compensation**: showing wages alone drops benefits — a large and growing share of pay.
- **Average vs. median**: productivity is an average; comparing it with *median* pay bundles two questions — how big is the pie, and how is it divided?
- **Deflator choice**: output (producer) prices and consumer prices can move differently — part of any "gap" is a price-index wedge.
- Also lurking: gross vs. net output (depreciation), and labor share vs. wage inequality — different questions, different charts.

Talent allocation and the size of the pie

- In 1960, 94% of U.S. doctors and lawyers were white men; by 2010, about 62%.
- Innate talent is unlikely to differ across groups — so 1960 implied a massive **misallocation of talent**.
- Hsieh, Hurst, Jones, and Klenow: falling barriers explain 20% to 40% of U.S. output-per-person growth, 1960–2010.
- Sandra Day O'Connor: third in her 1952 Stanford Law class — offered a job as a legal secretary.
- Opportunity and growth are **complements**, not a trade-off — the bridge to Part III.

Discussion

Four charts: productivity vs. average compensation; productivity vs. median wages; the labor share; household income after taxes and transfers. Which question does each one answer — and which chart would you show if you were a worker, a CEO, a union leader, or an economist?

Read before next class

- Stansbury and Summers (2018), “Productivity and Pay: Is the Link Broken?” (skim the figures).
- Hsieh, Hurst, Jones, and Klenow (2019), “The Allocation of Talent and U.S. Economic Growth,” *Econometrica* (introduction only).